Introduction

There is a lot of information available in the text format on the web. Reading that we can understand the sentiment about websites/products/brands in public tweets or in a public review.

It is hard to make the right decision without ever touching your product in real life.
So,People read reviews because they need help making informed decisions by others' experience.

The goal of this project is to understand and analyse the Online User Review Dataset with the help of different visualization techniques. These visualization techniques will help showcase various informative statistical trends which will provide us with insights about the Amazon Review system

However if you have the mission of getting and evaluating the users' feedback and you are going to live forever, maybe you have enough time to read all reviews in the Online Websites. Otherwise, teach the computer to do it for you will save your life!

The objective of the project is to extract, manipulate and create a good learning model to understand the user's sentiments classifying texts sent by a human being into Positive, Neutral and Negative.
Sentiment analysis is the process of analyzing text with the help of machine learning to identify the polarity of text.

There are mainly two approaches used for sentiment analysis. One method is to use the dictionary where each word is represented by a numerical value as polarity. The next method is machine learning, where statistical methods are employed to find out the vectorized value of a word by word embedding.

METHODOLOGY:

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As this project aims to perform sentiment classification of online product reviews using various Machine Learning classifiers

For web scraping we have used BeautifulSoup scraping product reviews

We have used Libraries such as numpy,
pandas for converting data into csv file,
Matplotlib, sklearn

Web scraping

Web scraping is the act of extracting or “scraping” data from a web page.

Python contains an amazing library called BeautifulSoup to allow web scraping.
Web scraping process is fully automated, done through a bot which we call the “Web Crawler”.
Web Crawlers are created using appropriate software like Python, with the BeautifulSoup and Scrapy libraries.

We have started **Scraping product details** and reviews from e-commerce website **Amazon**.

Generally, web scraping deals with extracting data automatically with the help of web crawlers. Web crawlers are scripts that connect to the world wide web using the HTTP protocol and allows us to fetch data in an automated manner.

First the targeted web page is “fetched” or downloaded. Next we the data is retrieved and parsed through into a suitable format.

We have used that to scrape product information and save the details in a CSV file.

Working :

First, we are going to import our required libraries.

Then we start with searching and collect all the names and asin numbers will take all the URL stored into a dataframe.

We will then feed all the URLs to our soup object.which will then extract relevant information from the given URL
based on the element id we provide it and

We balance the dataset accordingly without any inequality in the number of columns by adding NAN and keep balancing all the star ratings and reviews in it. So much so, there won't be any biassed data in the analysis process.

and At last all the data we collected will save it into our CSV file.

Preprocessing

Data cleaning is performed such as Lowercasing the data, Removing Punctuations, Removing Numbers, Removing extra space, Replacing the repetitions of punctuations, Removing Emojis, Removing emoticons, Removing Contractions

We have done Preprocessing steps like:

Handling Duplicates, Missing Values in data

>We have Dropped missing values in the “rating” column.

>“Title” and “customer_review” were combined and was kept under reviews column

>Rating which is in another format was converted to single digit format

>we dropped some columns which are not relevant

>we have added another column as EmotionScore and here Ratings greater than or equal to 3 have been categorised as “1” and less than 3 has been classified as “0”. Column Which is used in the process

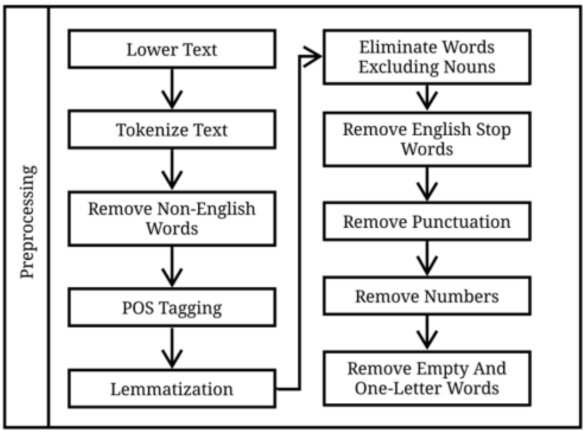

Fig. 2. Overall steps of preprocessing

Remove HTML tags from every single sample — because they won’t add meaningful value.

Remove punctuation signs — otherwise your model won’t understand that “good!” and “good” actually means the same thing.

DATA PREPARATION:
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The features like ‘reviewText’, ‘title’ and ‘rating’ as these were most useful and relevant for model building, prediction as well as abstraction
Here We only gave meaningful features for the model and not overfit on irrelevant noise
And applied NLP

NLP Techniques :

step-3(text Preprocessing):
As a next step we do preprocessing for scraped reviews and used some NLP techniques such as Stemming, stop-word removal and Lemmatization.

Text Preprocessing:

As a next step we would take all the scraped reviews and perform NLP techniques such as Stemming, stop-word removal and Lemmatization.

We have also perform de-duplication of our data and used some preprocessing before we go on further with analysis and making the prediction model.

Also done Preprocessing techniques like removing the html tags, Remove any punctuations or limited set of special characters like , or . or # etc.also removed some not alpha-numeric
Converted the words to lowercase

Remove Stop Words in it, Finally Porter Stemming is used.

After which we collect the data into dataframe.

Balancing the Datasets for Positive and Negative Sentiment. Since there is significantly more positive reviews than negative reviews, balancing the number of positive reviews to match the number of negative reviews is necessary

Feature Engineering

Creating Word Vectors Assigning Vector Representation for each word
Word2Vec tool.

here we will apply TF-IDF (term frequency Inverse Document Frequency) algorithm to convert string reviews into numeric vector. Each word count will be put in vector in place of words.

The process of converting the text reviews to a vector of intergers using one hot encoding. Deep learning models do not accept any textual inputs rather you need to feed the inputs as intgers or floats.
Word2vec is a method to efficiently create word embeddings by using a two-layer neural network.

(Used For DL)CountVectorizer simply counts the number of times a word appears in a document (using a bag-of-words approach)
(Used For ML)TF-IDF Vectorizer takes into account not only how many times a word appears in a document but also how important that word is to the whole corpus.
TF-IDF Vectorizer is used to convert words into numerical features and place weights on text based on importance and appearance frequency.

CountVectorizer is much simpler since it’s just a tool that converts a collection of text documents into a matrix of token counts, with no respect to the overall corpus.

Machine Learning

Multinomial Naive Bayes :

Naive bayes are good at text classification task like spam filtering, sentimental analysis, RS etc.

It is one of the most effective, widely used and a simple approach for text classification. In this approach, first
the prior probability of an entity being a class is calculated and the final probability is calculated by multiplying
the prior probability with the likelihood. The method is Naïve in the sense that it assumes every word in the text
to be independent. This assumption makes it easier to implement but less accurate

Logistic Regression :

Naive bayes are good at text classification task like spam filtering, sentimental analysis, RS etc.

Logistic Regression algorithm would perform relatively well as compared to the Naive Bayes algorithm. This can be due to the fact that the Logistic Regression algorithm doesn't make as many assumptions as that of the Naive Bayes algorithm.
LR only optimizes for the conditional likelihood, it has better performance than NB when the data disobeys the assumption (e.g. when two features are highly correlated).

logistic regression

The logistic regression model can be expressed as the conditional counterpart of the naive Bayes model.

$$\tilde{p}(x, y) = \psi(y) \prod_{n} \psi(x_i, y)$$

$$p(y|x) = \frac{1}{z(x)} \tilde{p}(x, y)$$

$$z(x) = \sum_{y} \tilde{p}(x, y).$$

Support Vector Machine (SVM) :

is a supervised learning algorithm based on vector theory.
The data is plotted in the form of vectors in space. Hyper-planes are used to make the decisions and classify the data points by keeping the different categories of data as far as possible from one another.

basic SVM requires much time while training the model in the context of data that is not linearly separable.

Additionally, the standard SVM classifier is not optimal for handling big data as it does not provide the correct results.

Random forest :

DeepLearning :

Deep Learning Model is created using neural networks. It has an Input layer, Hidden layer, and output layer.

The input layer takes the input, the hidden layer process these inputs using weights that can be fine-tuned during training, and then the model would give out the prediction that can be adjusted for every iteration to minimize the error.

(CNN) is a deep learning method used to analyze and map visual imagery.

Why ANN? :

ANN:(Artificial Neural Network)

is a group of multiple perceptrons or neurons at each layer.
The neural network with appropriate network structure can handle the correlation/dependence between input variables.

First, neural networks are data driven self-adaptive methods in that they can adjust themselves to the data without any explicit specification of functional or distributional form for the underlying model.
Second, they are universal functional approximates in that neural networks can approximate any function with arbitrary accuracy.

Since any classification procedure seeks a functional relationship between the group membership and the attributes of the object, accurate identification of this underlying function is doubtlessly important.

We have used three layers namely,
Sequential layer :
Sequential arranges the keras in sequential order/layer.
Data flows from one layer to another layer in a give order untill it reaches to final outputlayer.

Dense Layer:

Dropout Layer:
It is used for preforming of model.and it prevents overfitting.Drops the neurons from neural network

Results :

The best classifier was chosen to standardise the model to classify any product reviews in the future with promising outcomes.
we performed Cross-validation for selecting the machine learning model most appropriate model selection

As we have got better the outcomes with random forest. further we have made it standardise and built the model using Random forest.

Flask :

We have a flask App as front-End for our project. and built the website using html, css and javascript.

we have included pickle in it and we gave the output as gif's.

So here with-respect to the user input it classifies into different sentiment like positive, negative and neutral

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Used Tools :

For manipulation and analyze the data we have used the Pandas and Numpy libraries.

For Data Extraction the data from Amazon and Flipkart we have used Beautifulsoup webscraper respectively.

Used RE, Unidecode and NLTK libraries for Text Processing.

For lemmatization we will nltk.stem library and the module WordNetLemmatizer.

The nltk.corpus library we will use the module Stopwords to remove the english stop words.

For the Pipeline we are going to use Sklearn library, Base Estimator and Transformer Mixin modules.

For a nice and interactive plot we will use make_subplots, graph_objects and express modules from Ploty library.

For the Word Cloud we will use the module WordCloud from Matplotlib.pyplot.

To count the n grams for each sentiment we are going to use sklearn.feature_extraction.text librarie and CountVectorizer module.

For splitting train and test data we have used train_test_split module from sklearn.model_selection library.

For Embedding we will need Tensorflow. Keras library, Tokenizer and Sequence Pad_Sequences modules for applying the GloVe method.

For Balancing we will use from imblearn library and RandomOverSampler module.

For Feature selection, we are going to use Sklearn feature selection library and F_classif ,Select K Best modules

For Evaluation Metrics we use functions to calculate manually the Precision, Recall and F1 score, using backends module from tensorflow Keras library.

For the Neural Network we are going to use tensorflow.Keras library and modules Model, Dense, LSTM, Input and Embedding.

For plot the model we will use from plot_model module from tensorflow.keras.utils library.

Again to plot the history of the model on the training set, on the validation set and the on the final prediction labels we are going to use matplotlib library.